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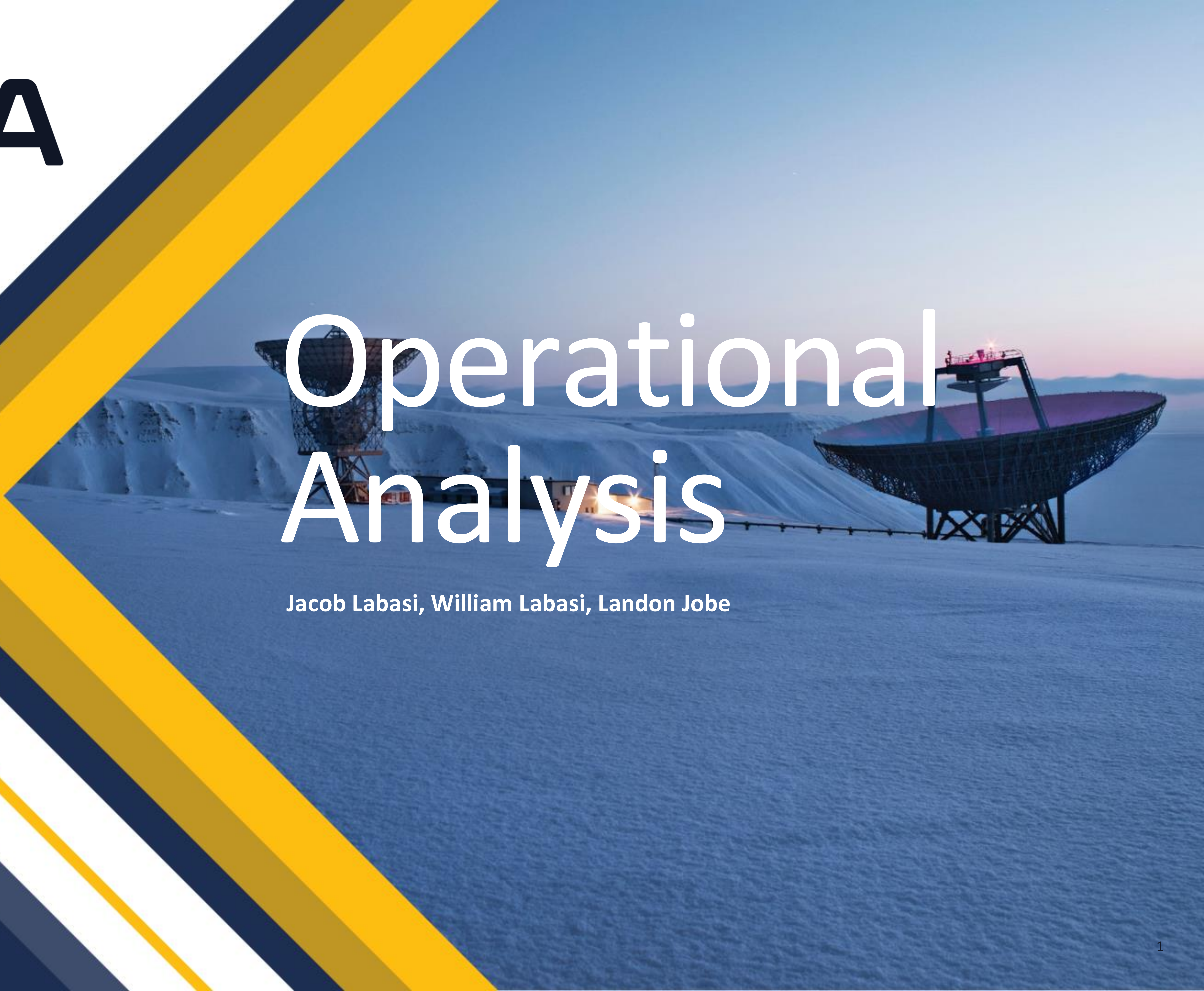

Operational Analysis

Jacob Labasi, William Labasi, Landon Jobe

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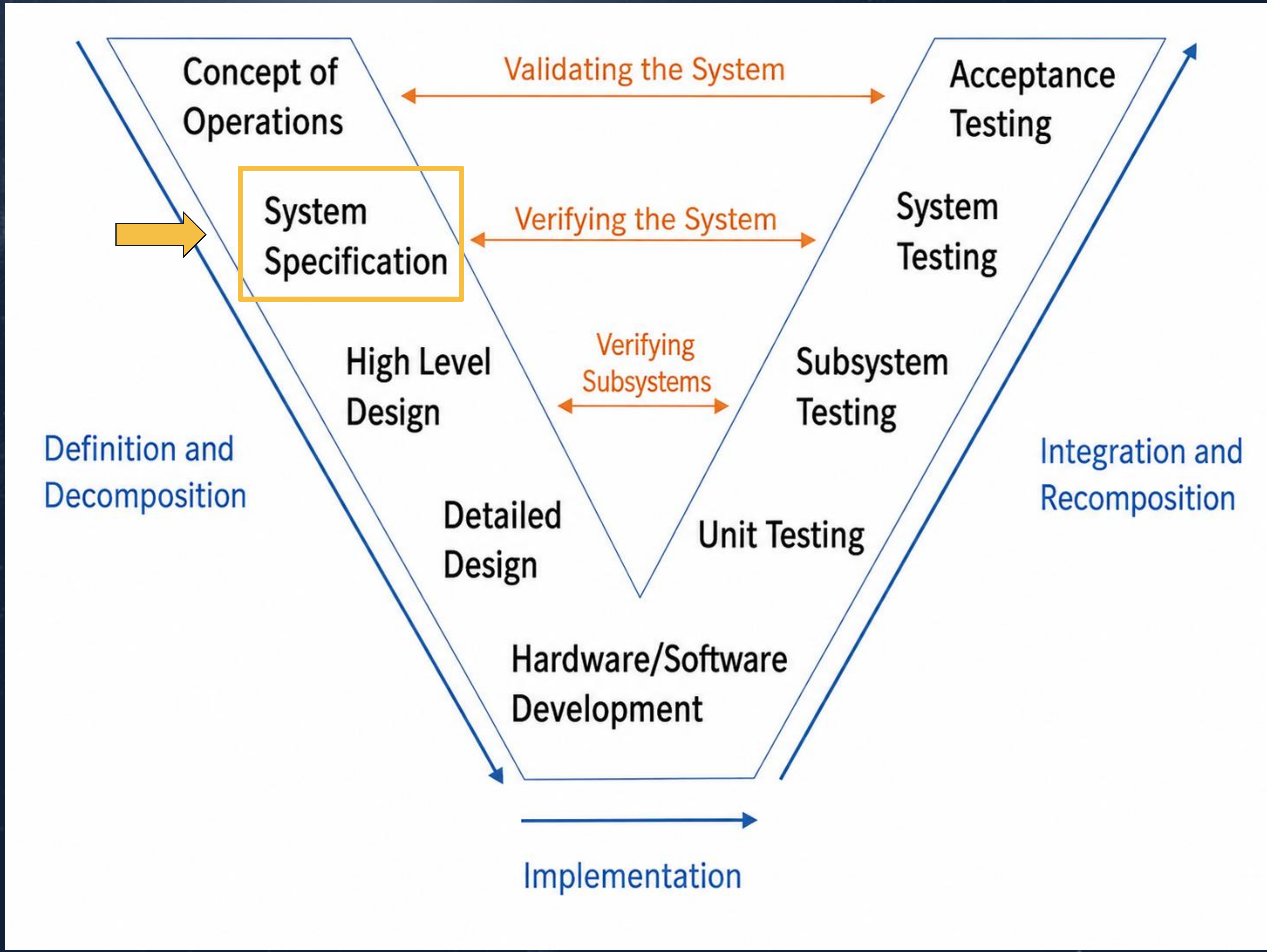
Phase 2: Operational Analysis

V-Diagram Update



Goal and Deliverables

Goal: Develop a complete understanding of how the Digital Twin will be used within Auria's operational environment.

Deliverables: Stakehodler analysis, requirements, data flow diagrams, CONOPS, mission definition, As-Is and To-Be states, inputs and outputs

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CONOPS (Who, What, When, How, Why)

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Why? The Mission

The Mission

Problem: Space operations generate large volumes of orbital, telemetry, environmental, and mission data that are often distributed across multiple systems and tools. Without an integrated Digital Twin framework, operators face challenges in maintaining situational awareness, monitoring system health, predicting future system behavior, assessing operational risks, and making timely mission decisions.

Solution: Develop a scalable Digital Twin framework that models, monitors, simulates, and predicts the behavior of a satellite-based space operations system to enhance space domain awareness, mission readiness, system health monitoring, collision avoidance, and operational decision-making across Auria's space mission environment.

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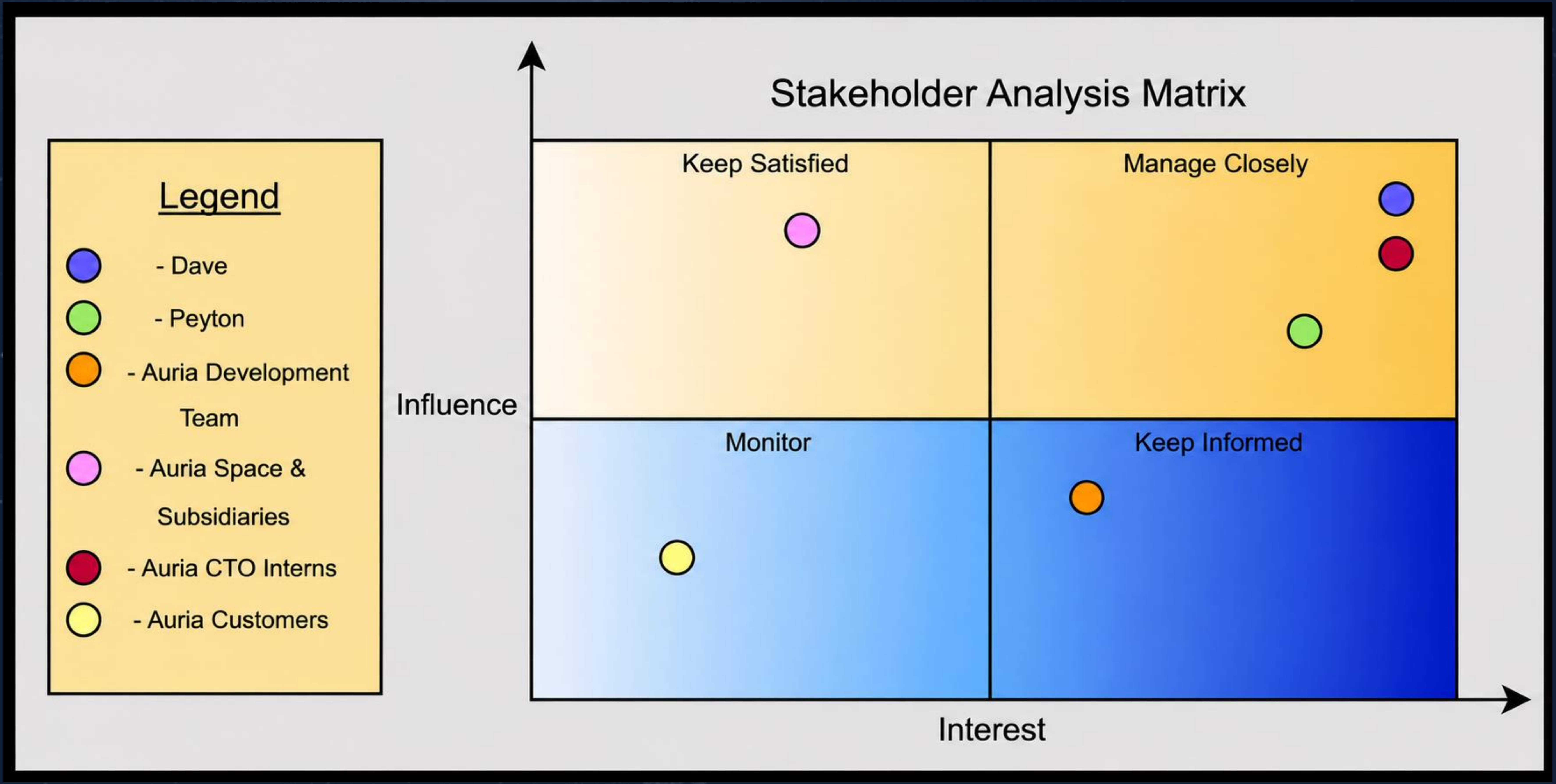
Who? Stakeholders

Stakeholder Analysis



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When? Future (To-Be) and Current (As-Is) State

Future and Current State

Current: Space operations data is scattered across various tools, databases, and systems. Operators manually collect information from satellite telemetry, tracking databases, mission planning tools, and communication systems to assess satellite health, monitor orbital activity, identify threats, and support mission planning. This demands significant operator effort and limits predictive analysis and situational awareness.

Future: The Digital Twin offers a unified virtual model of the satellite system that continuously integrates telemetry, orbital, environmental, and mission data. Operators monitor system status, predict behavior, assess collision risks, simulate scenarios, and evaluate mission decisions within a single environment.

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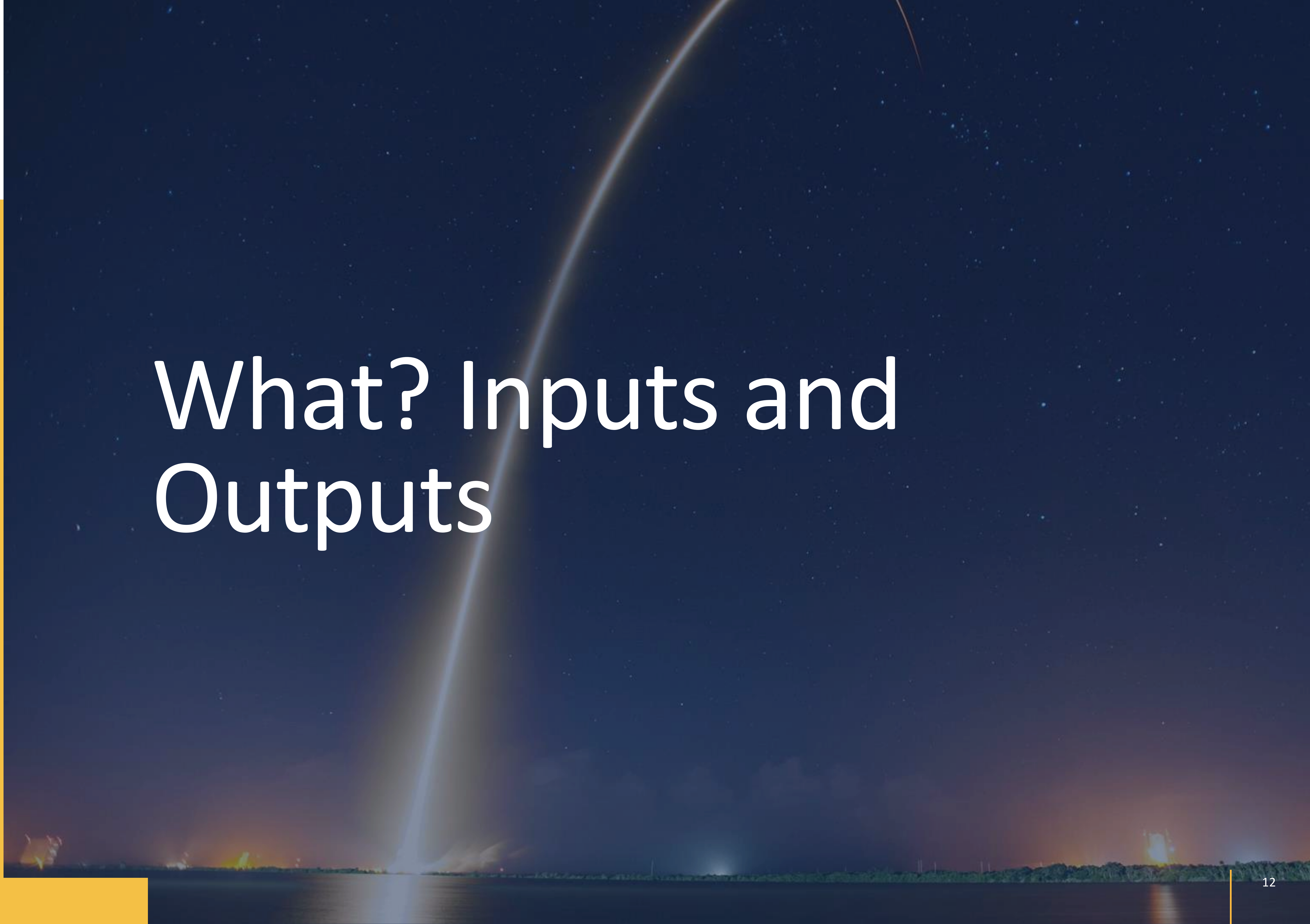

What? Inputs and Outputs

Revised Inputs/Outputs

Inputs

- Telemetry
- Space Surveillance Tracking Data
- Environmental Data
- Mission objectives
- Command History
- Command and Control

Outputs

- Orbital Data: Give visual representation of latitude, longitude, altitude, velocity and time of the satellite.
- Collision Risk: Use (SSTD, orbital, environmental) data to provide collision risks assessments by utilizing machine learning.
- Health Status: Use telemetry data and command history to provide visual system health status of the satellite.
- Operator Recommendations (COAs): Use all collected data/inputs to produce system recommended operator upgrades, planning, etc.
- Object Identification: Red vs. Blue

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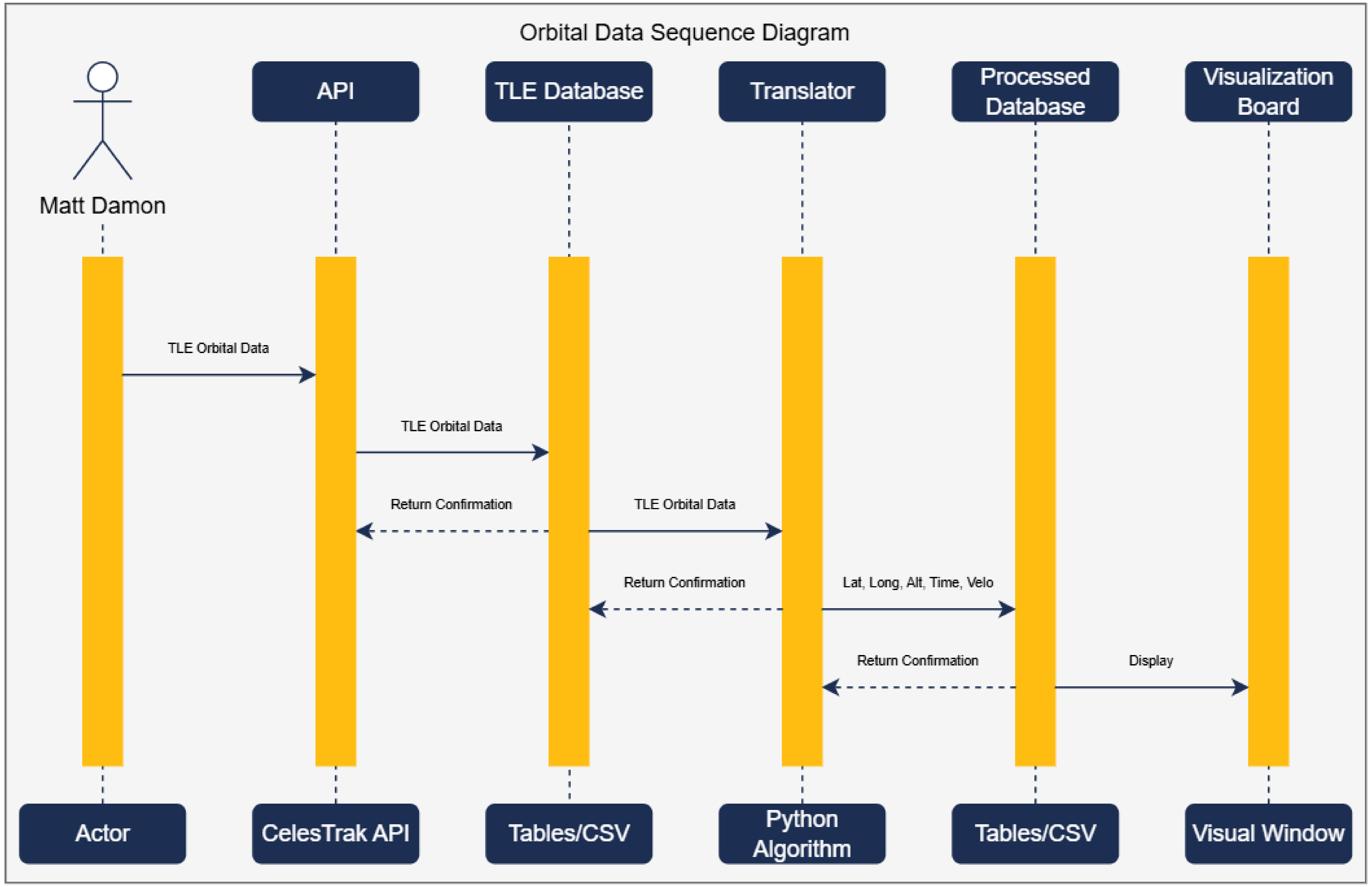
How? Data Flow Diagrams (3/5)

Orbital Data



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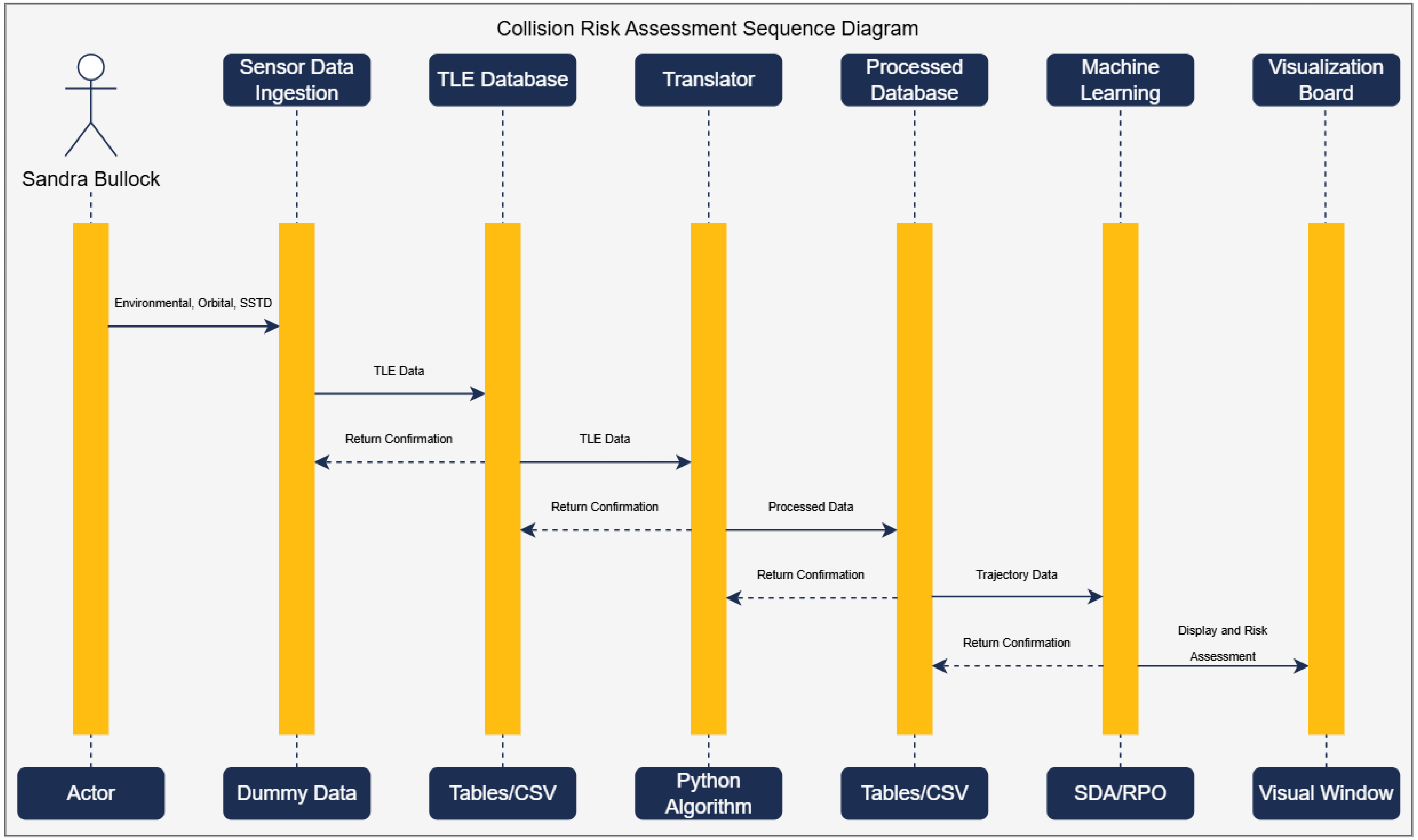
Collision Risk Assessment



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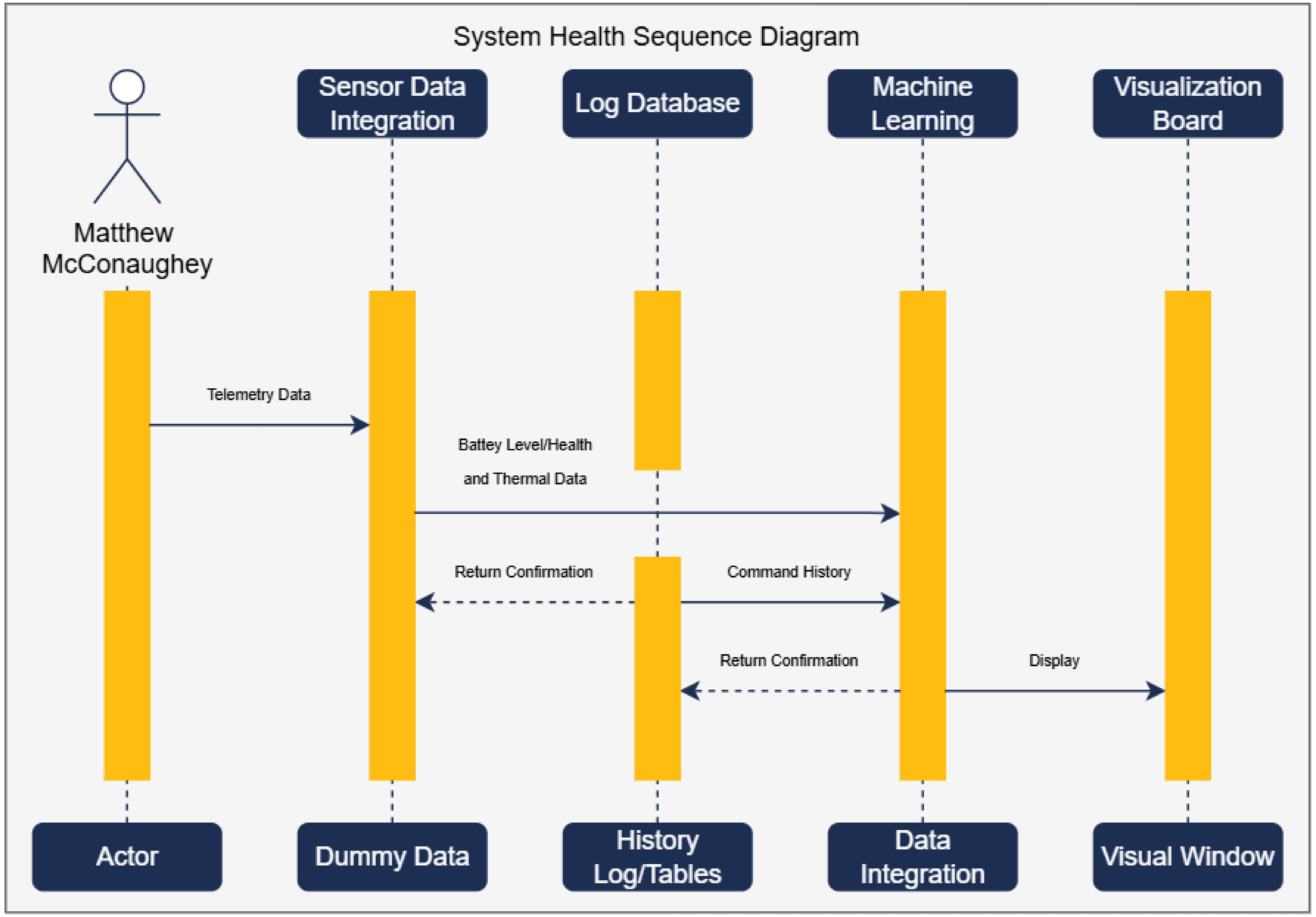
System Health



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Requirements

Requirements

0	Create a Digital Twin (DT) Framework
REQ-CORE-1.0	
1.1	***The framework shall be able to process inputs with expected outputs***
1.1.1	**The framework shall use tools for implimentation**
1.1.1.1	The DT shall be coded within Python
1.1.1.2	The DT shall be created in GitLab
1.1.1.3	The DT shall be implemented by Cerux for C3
1.1.1.4	The DT shall be contained in Docker
1.1.1.4.1	The DT shall use Kubernetes to run Docker
1.1.1.5	Data shall be developed within amazon workspace
1.1.1.6	The DT shall contain a visualization board
1.1.1.7	The DT shall simulate orbital mechanics
1.1.1.8	All data shall be stored within a database
1.1.2	**The framework shall provide command and control data**
1.1.2.1	C2 data shall provide movement commands
1.1.2.1.1	C2 data shall provide rotational movement
1.1.2.1.2	C2 data shall reposition the system
1.1.3	**The framework shall simulate environmental impacts to system**
1.1.3.1	The framework shall simulate weather conditions
1.1.3.2	The framework shall simulate unknown objects
1.1.3.2.1	Simulation data shall consider unidentified technologies
1.1.4	**All data shall be in metric units**

REQ-APP-1.0	
1.1	***The framework shall be applicable for different use cases***
1.1.1	**The framework shall simulate mission readiness of combat systems**
1.1.1.1	The framework shall implement cycbersecurity testing
1.1.1.2	The framework shall simulate cycbersecurity vulnerability
1.1.1.3	The framework shall highlight future technology development
1.1.2	**The framework shall monitor environmental effects on different systems**
1.1.2.1	Sensors shall detect radiation
1.1.2.2	Sensors shall detect geomagnetic storms
1.1.2.3	Sensors shall detect atmospheric disturbances
1.1.2.4	Signal and network resiliance for environmental conditions
1.1.3	**The framework shall provide operator training for system use**
1.1.3.1	Simulated ambiguous events for operator mission rehearsal
1.1.3.2	Simulated multidomain training for operator rehearsal
1.1.3.3	Simulated C2 training for operator rehearsal
1.1.3.3.1	Simulated network training for operator rehearsal
1.1.3.3.2	Simulated communication training for operator rehearsal
1.1.3.3.3	Shall be applicable to Forgepoint operator training
1.1.4	**The framework shall provide C3 system development**
1.1.4.1	The framework shall contain visual health monitoring indicators for the system
1.1.4.1.1	Green visual indicators shall correspond to normally operating
1.1.4.1.2	Yellow visual indicators shall correspond to at risk to fail systems
1.1.4.1.3	Orange visual indicators shall correspond to soon to fail systems
1.1.4.1.4	Red visual indicators shall correspond to failed systems

REQ-INT-1.0	
1.1	***The framework shall integrate with Auria applications***
1.1.1	**The DT receives master satellite scheduling**
1.1.1.1	DT shall emit a timestamp
1.1.1.2	DT shall emit event_ID
1.1.2	**The framework shall process telemetry data to determine system health**
1.1.2.1	The telemetry data shall contain orientation data
1.1.2.2	The telemetry data shall contain power levels
1.1.2.3	The telemetry data shall contain power health
1.1.2.4	The telemetry data shall contain temperature data
1.1.3	**The DT shall contain different options for data compatibility**
1.1.3.1	Data shall be consumed to TeRMS
1.1.3.2	Data shall be consumed by MFPA

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What's Next?

Upcoming

- June 23rd: meeting with Ken Center to discuss collision risk assessment in relation to SDA and RPO
- Next week: potential meeting with Patty Bonvallet to discuss SDA, RPO, ForgePoint and BORG
- Finalize data sequence diagrams and begin block diagram
- Mission objective sequence diagram
- Meet *Dave....*